

MA XUCHU

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EDUCATION

Xi'an Jiaotong University, Xi'an, China

Master of Electrical Engineering | GPA: 3.59/4.0 (Rank 2/11)

Sep 2021 – Jul 2024

Southwest Jiaotong University, Chengdu, China

Bachelor of Electrical Engineering and Automation | GPA: 3.65/4.0 (Rank 8/98)

Sep 2017 – Jul 2021

RESEARCH EXPERIENCE

Deep Learning Methods for Transformer Condition Prediction and Fault Diagnosis

Based on Dissolved Gas Analysis

Jul 2022 – Jul 2024

Xi'an Jiaotong University | Main Researcher

- Developed a CBAM-SIREN-LSTM model for forecasting dissolved gas concentrations, improving MAPE to 2.86% and RMSE to 1.87, outperforming LSTM, CNN-LSTM, and CBAM-LSTM baselines.
- Designed a CA-CNN fault diagnosis model with K-means SMOTE balancing, achieving 93.26% validation accuracy and 87.16% test accuracy, substantially higher than traditional DGA methods (42.46–67.8%).
- Strengthened expertise in deep learning architectures, time-series forecasting, feature engineering, and data-driven fault diagnosis using Python (PyTorch, NumPy, Pandas).

Modeling the Maximum Allowable Laying Length of High-Voltage Cables

Jul 2021 – Dec 2022

Xi'an Jiaotong University | Researcher

- Simulated electrical limits using ATP-EMTP (maximum 2810 m) and computed mechanical constraints via MATLAB-based force modeling (1700 m in the case study).
- Enhanced skills in numerical simulation.

Parallel Control Technology of Metro Auxiliary Inverter

Jul 2020 – Jun 2021

Southwest Jiaotong University | Undergraduate Thesis Project

- Proposed an SVPWM-based voltage-current dual-loop control for single inverter operation and a resistive droop control method for parallel systems.
- Built and verified inverter control models in MATLAB/Simulink, confirming reliable voltage and current control under various load conditions.
- Gained practical understanding of inverter topologies, control strategy design.

PUBLICATIONS

Contributed to the paper “Analysis of the optimal parameters for serpentine laying of high-voltage cables considering the influence of various factors,” presented at the *3rd International Conference on Electrical, Electronic Information and Communication Engineering (EEICE 2022)*, Guilin, China.

PROFESSIONAL EXPERIENCE

Xi'an Power Supply Company, State Grid Shaanxi Electric Power Co., Ltd.

Aug 2024 – Present

Engineer, Transmission Operation and Maintenance Center

Xi'an, China

- Conducted operation and maintenance work for high-voltage transmission lines, gaining practical experience in power grid infrastructure, field inspection workflows, and equipment condition assessment.
- Participated in UAV-based autonomous inspection tasks, including transmission-line point cloud data acquisition and UAV flight route planning.

AWARDS AND HONORS

Special Scholarship for Academic Excellence (2022, 2023)

Excellent Postgraduate Cadre (2023)

Outstanding Graduate, Southwest Jiaotong University (2021)

First Prize, CSEE Cup National University Student Mathematical Modeling Competition (2018)

SKILLS

Programming: Python (PyTorch, NumPy, Pandas, Matplotlib, PySide6), C++, MATLAB

Language: English (TOEFL iBT: 91 [R24, L27, S17, W23])